

Reider's Theorem, More and Less

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Let S be a smooth projective surface over $\mathbb{C}$.

1 Strategy

Going back to the ideas of [1], we will prove variants of Reider's theorem by finding suitable Bridgeland stability conditions.

Specifically, to show a line bundle $\omega_S(C)$ is d -very ample, we will get the vanishing

$$H^1(S, \mathcal{I}_Z \otimes \omega_S(C)) = 0,$$

as Z ranges over all length d subschemes. This cohomology group can be rewritten (with the help of Serre duality) as

$$\begin{aligned} \mathrm{Ext}_{\mathcal{O}_S}^1(\mathcal{O}_S(-C), \mathcal{I}_Z \otimes \omega_S) &= \mathrm{Ext}_{\mathcal{O}_S}^1(\mathcal{I}_Z, \mathcal{O}_S(-C))^\vee \\ &= \mathrm{Hom}_{D^b(S)}(\mathcal{I}_Z, \mathcal{O}_S(-C)[1])^\vee. \end{aligned}$$

To preclude any nonzero morphism $\mathcal{I}_Z \rightarrow \mathcal{O}_S(-C)[1]$ in the derived category, we will then find a Bridgeland stability condition with $\mathcal{I}_Z, \mathcal{O}_S(-C)[1]$ both (semi)stable but such that $\mathcal{I}_Z$ has greater slope.

Can we use this strategy to get Koszul cohomology to vanish by phrasing it as vanishing of cohomology of kernel bundles as in Ein-Lazarsfeld Theorem 5.2.5? This could work towards projective normality as in page 143?

2 Stability

2.1 Review

We use notation from [2], which includes a choice of smooth coordinates for a “slice” (of a chart) of the stability manifold. Specifically, we get a stability condition for each (x, y) inside the parabola $x > \frac{1}{2}y^2$, which is the data of a central charge $Z_{(x,y)}$ and a heart of a bounded t-structure $\mathcal{A}_{-y}$.

$$Z_{(x,y)}(E) = (-\mathrm{ch}_2(E), c_1(E) \cdot H) + (\mathrm{rk} E)(x, y)$$

$$\mathcal{A}_{-y} = \{E \in D^b(S) \mid -\mu_{\min}(\mathcal{H}^0(E)) < y \leq -\mu_{\max}(\mathcal{H}^{-1}(E)), \mathcal{H}^i(E) = 0 \text{ for } i \neq -1, 0\}$$

We recall here that as we increase y , we make the following trade:

- We *gain* H -semistable coherent sheaves with low H -slope, like $\mathcal{O}_S(-nC)$
- We *lose* shifts of H -semistable coherent sheaves with low H -slope, like $\mathcal{O}_S(-nC)[1]$.

In particular, an Mumford H -stable coherent sheaf $\mathcal{F}$ appears in the category when $y > -\mu(\mathcal{F})$.

Note that when $\mathcal{F}$ is *not* H -semistable, we have to wait a while after we lose $\mathcal{F}[1]$ before we gain $\mathcal{F}$. For instance, on $\mathbb{P}^2$ for $\mathcal{F} = \mathcal{O}_{\mathbb{P}^2}(-1) \oplus \mathcal{O}_{\mathbb{P}^2}(1)$, we have Harder-Narasimhan filtration

$$0 \leq \mathcal{O}_{\mathbb{P}^2}(1) \leq \mathcal{O}_{\mathbb{P}^2}(-1) \oplus \mathcal{O}_{\mathbb{P}^2}(1),$$

so has maximal H -slope $\mu_{\max}(\mathcal{F}) = 1$ and minimal slope $\mu_{\min}(\mathcal{F}) = -1$. Therefore, $\mathcal{F}[1]$ leaves the category once $y > -(1)$, but $\mathcal{F}$ only joins the category once $y > -(-1)$.

Also important is a description of (numerical) walls, which are lines through *points of indeterminacy* $I(E)$ of objects $E \in D^b(S)$, which are points $I(E) \in \mathbb{R}^2$ such that $Z_{I(E)}(E) = 0$. These have coordinates

$$I(E) = \left(\frac{\text{ch}_2 E}{\text{rk} E}, -\mu E \right).$$

Any H -semistable coherent sheaf $\mathcal{F}$ (or a shift, which has identical point of indeterminacy) has $I(\mathcal{F})$ outside or on the parabola—this a version of the Bogomolov inequality. Alternatively, we can rephrase this argument as follows: if $\mathcal{F}$ is H -semistable, then $\mathcal{F} \in \mathcal{A}_{-\mu\mathcal{F}}$ so it is in the category at $I(\mathcal{F})$, so (x, y) cannot give a stability condition because the central charge sends an object to 0. Thus, (x, y) is outside the parabola.

When $\mathcal{F}$ fails to be semistable, like our example $\mathcal{F} = \mathcal{O}_{\mathbb{P}^2}(-1) \oplus \mathcal{O}_{\mathbb{P}^2}(1)$, we might have $I(\mathcal{F})$ inside the parabola, but it will occur at a y such that $\mathcal{F} \notin \mathcal{A}_{-y}$. For example, for this $\mathcal{F}$,

$$I(\mathcal{F}) = \left(\frac{\text{ch}_2(\mathcal{O}_{\mathbb{P}^2}(-1)) + \text{ch}_2(\mathcal{O}_{\mathbb{P}^2}(1))}{2}, \frac{-(-1) - (1)}{2} \right) = \left(\frac{1}{2}, 0 \right),$$

and indeed, $\mathcal{O}_{\mathbb{P}^2}(-1) \oplus \mathcal{O}_{\mathbb{P}^2}(1) \notin \mathcal{A}_0$ as discussed earlier.

2.2 $\mathcal{O}_S(-C)$ and Bogomolov-sharp Sheaves

Now, we analyze when $\mathcal{O}_S(-C)[1]$ is (semi)stable. First, we describe how, for an H -semistable torsion free coherent sheaf F , might F or $F[1]$ be destabilized. In particular, we need to know what a sub/quotient looks like in one of our hearts $\mathcal{A}_{-y}$.

Lemma 2.1. Let F be as above. Then,

- (1) If $K \leq F$ in $\mathcal{A}_{-y}$ with cokernel Q , then, by the long exact sequence in $\text{Coh}(S)$ cohomology,

$$0 \rightarrow \mathcal{H}^{-1}(K) \rightarrow 0 \rightarrow \mathcal{H}^{-1}(Q) \rightarrow \mathcal{H}^0(K) \rightarrow F \rightarrow \mathcal{H}^0(Q) \rightarrow 0$$

K is itself a sheaf with $\mu_{\min}(K) > -y$ (and torsion free, being an extension of $\mathcal{H}^{-1}(Q)$ and a subsheaf of F).

- (2) If $F[1] \twoheadrightarrow Q$ is a surjection in $\mathcal{A}_{-y}$ with kernel K , then, by the long exact sequence in $\text{Coh}(S)$ cohomology,

$$0 \rightarrow \mathcal{H}^{-1}(K) \rightarrow F \rightarrow \mathcal{H}^{-1}(Q) \rightarrow \mathcal{H}^0(K) \rightarrow 0 \rightarrow \mathcal{H}^0(Q) \rightarrow 0,$$

$Q[-1]$ is itself a sheaf with $\mu_{\max}(Q[-1]) \leq -y$ (so torsion free). **check this... doesn't need F torsion free?**

Lemma 2.2. Let F be H -stable torsion free. Suppose there exists (x, y) in the parabola such that F is semistabilized by some proper subobject $E \xrightarrow{\varphi} F$ in $\mathcal{A}_{-y}$. Then, (x, y) is above $I(E)$ above $I(F)$, i.e. $y > -\mu(E) > -\mu(F)$.

Proof. First, assume $\text{rk} F > 0$, since otherwise the lemma is automatic. Continuing, since $E, F \in \mathcal{A}_{-y}$, we know

$$y > -\mu(F), -\mu_{\min}(E) \geq -\mu(E),$$

so at least (x, y) is above $I(E), I(F)$.

For $E \xrightarrow{\varphi} F$ to be a monomorphism in $\mathcal{A}_{-y}$, need the cone $C(\varphi) \in \mathcal{A}_{-y}$, which in particular means is the requirement that

$$\mu_{\max}(\mathcal{H}^{-1}(C(\varphi))) = \mu_{\max}(\ker_{\text{Coh}(S)} \varphi) \leq -y$$

Denote this kernel by K . If $K = 0$, then E injects in F in $\text{Coh}(S)$ and we win by H -stability of F . Otherwise, $K \neq 0$, is torsion free (by virtue of finite slope), so has positive rank. On the other extreme, we also cannot have $K = E$, since then

$$\mu(E) \leq -y, \quad \mu(E) > -y.$$

So, E/K is nonzero. Now, we have short exact sequence in $\text{Coh}(S)$

$$0 \rightarrow K \rightarrow E \rightarrow E/K \rightarrow 0,$$

which gives

$$H \cdot c_1(E) = H \cdot c_1(K) + H \cdot c_1(E/K), \quad (\star)$$

and since $E/K \leq F$ is a nonzero subobject in $\text{Coh}(S)$ with F torsion free, we know E/K is torsion free and

$$1 \leq \text{rk} E - \text{rk} K \leq \text{rk} F, \quad \mu(E/K) < \mu(F).$$

In particular, $\text{rk} E \geq 1$, so it has finite H -slope, and we can use $(\star)$ to solve for $\mu(E)$, giving us the equation

$$\mu(E) = \frac{\text{rk}(K)\mu(K)}{\text{rk} E} + \frac{\text{rk}(E/K)\mu(E/K)}{\text{rk} E}.$$

Plugging in our other inequalities, in particular using

$$\mu(K) \leq -y < \mu(F)$$

we get

$$\begin{aligned} \mu(E) &< \frac{\text{rk}(K)}{\text{rk}(E)}\mu(F) + \frac{\text{rk}(E) - \text{rk}(K)}{\text{rk}(E)}\mu(F) \\ &= \mu(F). \end{aligned}$$

□

Lemma 2.3. With the same assumption as lemma 2.2, but also assuming E is (x, y) -semistable, $I(E)$ is outside the parabola and on the same side of the parabola as $I(F)$.

Proof. $I(E)$ is outside of the parabola by a variant of the Bogomolov inequality for (x, y) semistable objects (add reference/collect Bayer-Macri-Toda theorem here too). Then (x, y) above $I(E)$ above $I(F)$ as in lemma 2.2 implies the result. $\square$

Lemma 2.4. Let F be an H -semistable torsion free sheaf. Then, F is $(x, -\mu(F) + \varepsilon)$ semistable for $x \gg 0$ and $0 < \varepsilon \ll 1$.

Proof. prove or cite this!! probably also need more in my applications to be careful... $\square$

Lemma 2.5. Let F be (x, y) -semistable such that in every neighborhood of $(x, y) \in \text{Stab}(S)$, there exists an (x', y') such that F becomes unstable. Then, we can choose an (x, y) -semistable object E with the same (x, y) slope as F , such that E is a subobject in $\mathcal{A}_{-y}$ and such that $E \leq F$ destabilizes F in some (x', y') in any small neighborhood.

Furthermore, the quotient F/E in $\mathcal{A}_{-y}$ is also (x, y) -semistable (this follows quickly).

Proof. state this lemma more carefully... do I need a requirement on the slope so that I can have a constant category somehow? Choose a sequence (x_n, y_n) converging to (x, y) , such that F is (x_n, y_n) unstable for all $n \in \mathbb{N}$, evidenced by a subobject $E_n \hookrightarrow F$ in $\mathcal{A}_{-y_n}$. Even more, choose E_n to be the first factor of the (x_n, y_n) Harder-Narasimhan filtration, so that E_n is (x_n, y_n) semistable (and so this semistability spreads out in a neighborhood).

Eventually, E_n becomes a subobject in $\mathcal{A}_{-y}$? Local finiteness of walls? Furthermore, the $\nu_{(x, y)}(E_n) \leq \nu_{(x, y)}(E)$ by virtue of the (x, y) -semistability of E , but for $n \gg 0$ we must get equality because $\nu_{(x, y)}(E_n)$ is continuous in (x, y) , and the inequality is reversed (and strict) at (x_n, y_n) . fix this later!!! $\square$

Definition 2.6. As in lemma 2.5, we say

- the wall through $I(F)$ and (x, y) is a **destabilizing wall** for F ,
- the subobject $E \hookrightarrow F$ in $\mathcal{A}_{-y}$ is a **witness** for the wall.

Now, we apply these lemmas to get the stability of $\mathcal{O}_S(-C)[1]$ and friends, using the observation that

$$I(\mathcal{O}_S(-C)[1]) = \left(\frac{C^2}{2}, \|C\| \right)$$

is exactly on the parabola. In this sense, $\mathcal{O}_S(-C)[1]$ is *Bogomolov-sharp* (the inequality became an equality), and we will show such Bogomolov-sharp object are stable in the following sense.

Proposition 2.7. Let $F \in \text{Coh}(S)$ be H -stable and torsion free such that $I(F)$ lands on the parabola $x = \frac{1}{2}y^2$. Then,

- (1) F is (x, y) semistable for all (x, y) in the parabola with $y > -\mu(F)$.

(2) $F[1]$ is (x, y) semistable for all (x, y) in the parabola with $y \leq -\mu(F)$.

Proof. We show (1). By H -stability, F will be $(x, -\mu(F) + \varepsilon)$ semistable for $x \gg 0$ and $0 < \varepsilon \ll 1$ as in lemma 2.4, and F will *remain* (x, y) -semistable until just after a destabilizing wall for F counterclockwise to the positive x -axis. Choose a witness

$$0 \rightarrow E \rightarrow F \rightarrow F/E \rightarrow 0$$

for this wall, as in lemma 2.5. By lemma 2.2 and 2.3, $I(F), I(E), (x, y)$ form a positive slope line with increasing heights in order with $I(E)$ outside the parabola, but this is impossible because the line segment $I(F)$ to (x, y) is entirely inside the parabola. Thus, $I(F)$ couldn't have been destabilized. $\square$

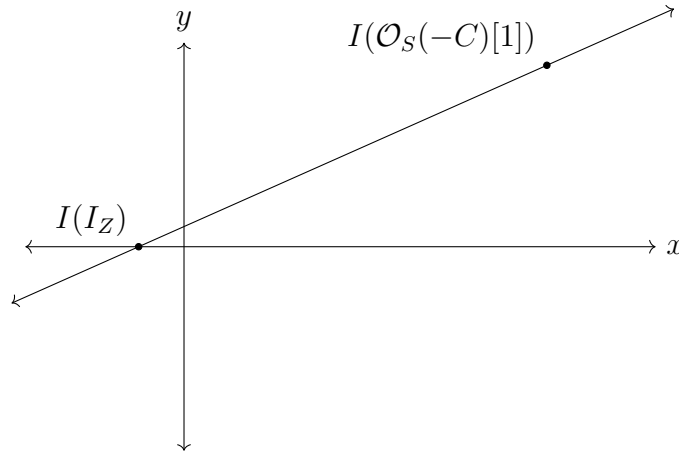
TODO: show (2) in a parallel way... might need to expand the lemmas though.

2.3 Ideal Sheaves

Unlike $\mathcal{O}_S(-C)[1]$, an ideal sheaf $\mathcal{I}_Z$ for a length d subscheme will not always be stable. However, we will establish numerical criteria to guarantee stability of the $\mathcal{I}_Z$ long enough to find (x, y) where $\mathcal{I}_Z$ has bigger (x, y) slope than $\mathcal{O}_S(-C)[1]$.

By openness of stability, we just need to guarantee $\mathcal{I}_Z$ is stable at a point on the line through $I(\mathcal{I}_Z)$ and $I(\mathcal{O}_S(-C)[1])$ intersected with the parabola. By lemma 2.4, since $\mathcal{I}_Z$ is H -stable torsion free, we at least know $\mathcal{I}_Z$ is semistable at $(x, -\mu(F) + \varepsilon)$ for $x \gg 0$ and $0 < \varepsilon \ll 1$, i.e., $\mathcal{I}_Z$ is stable in some circular sector bounded by the positive x -axis and some positive slope ray.

So, we just need to show that F *remains* stable through the ray from $I(\mathcal{I}_Z)$ to $I(\mathcal{O}_S(-C)[1])$.



Now, for a point (x, y) on a destabilizing wall for $\mathcal{I}_Z$, choose a witness $F \xrightarrow{\varphi} \mathcal{I}_Z$ (again as in lemma 2.5). We will refer to $\text{rk} F$ as the *rank of the wall* for $\mathcal{I}_Z$ through $I(F)$. We want to establish numerical conditions on (S, C) to prevent any walls that occur up to the $\mathcal{I}_Z, \mathcal{O}_S(-C)[1]$ wall. Our target wall has slope (in our coordinates for $\text{Stab}(S)$)

$$m = \frac{\|C\|}{\frac{C^2}{2} + d} = \frac{2\|C\|}{C^2 + 2d},$$

so we want to guarantee that any other wall through I_Z has greater slope.

2.3.1 Higher Rank Walls

Suppose $\text{rk} F = r > 1$. As in [2], we show that this implies the wall W has too great a “height”, i.e., showing that the ratio of the y coordinates of the top intersection and bottom intersection of $W \cap \{x = \frac{1}{2}y^2\}$ is too large.

Computation 2.8. For comparison, we compute this height ratio for the $I_Z, \mathcal{O}_S(-C)[1]$ wall. The top intersection has y coordinate $\|C\|$. The bottom intersection has is one of the solutions to the equations

$$2x = y^2, \quad y = m(x + d),$$

which has y -coordinate which is a root of

$$\begin{aligned} y &= m \left(\frac{1}{2}y^2 \right) + md \\ 0 &= y^2 - \frac{2}{m}y + 2d. \end{aligned}$$

Since one root is $\|C\|$, the other root is $\frac{2d}{\|C\|}$. Thus, the height ratio is $\frac{C^2}{2d}$ (or potentially the reciprocal if $\|C\|$ isn't sufficiently large—but we will eventually require $\|C\|$ to be large enough so that $\|C\|$ is the bigger height).

Now, we bound the height ratio for a rank r wall. By Lemma 2.3, $I(F)$ is positioned between $I(I_Z)$ and the parabola, with positive y -value. Writing E for the cone, so that we have exact sequence

$$0 \rightarrow F \rightarrow I_Z \rightarrow E \rightarrow 0 \quad (\star)$$

in $\mathcal{A}_{-y}$ (where (x, y) is on the destabilizing wall). Since F, I_Z are both (x, y) -semistable of the same slope, so is E . In this setup, $I(I_Z)$, $I(F)$, (x, y) , and $I(E)$ are all on the same line, and we at least know $I(I_Z)$ below $I(F)$ below (x, y) . Furthermore, by (x, y) stability, $I(F), I(E)$ are outside the parabola.

Proposition 2.9. $I(E)$ is above (x, y) .

Proof. We have to show $\mu(E) < -y$. We know that $E \in D^b(S)$ has cohomology sheaves

$$\mathcal{H}^{-1}(E) = \ker_{\text{Coh}(S)} \varphi, \quad \mathcal{H}^0(E) = \text{coker}_{\text{Coh}(S)} \varphi.$$

Since $\varphi \neq 0$ for φ to be a monomorphism, we in fact get

$$\text{rk}(\text{coker}_{\text{Coh}(S)} \varphi) = 0$$

since $\text{coker}_{\text{Coh}(S)} \varphi = I_Z / \text{im}_{\text{Coh}(S)} \varphi$ is a quotient of rank 1 sheaves, where $\text{im}_{\text{Coh}(S)} \varphi$ is rank 1 because it is a nonzero subsheaf of a torsion free sheaf. Thus, $\text{coker}_{\text{Coh}(S)} \varphi$ is a torsion sheaf, and in particular

$$c_1(\text{coker}_{\text{Coh}(S)} \varphi) \geq 0$$

is effective, since $c_1 = \text{char}_1$ can be computed by taking lengths of $\text{coker}_{\text{Coh}(S)}\varphi$ at generic points of components of the support (i.e., filter $\text{coker}_{\text{Coh}(S)}\varphi$ by structure sheaves of curves and points by analyzing associated primes/points).

On the other hand, $\text{rk}(\ker_{\text{Coh}(S)}\varphi) = -\text{rk}F$, since $\text{rk}F$ is the alternating sum of the ranks of its cohomologies.

Thus, we can compute

$$\mu(E) = \mu(\ker_{\text{Coh}(S)}\varphi) + \frac{H \cdot c_1(\text{coker}_{\text{Coh}(S)}\varphi)}{\text{rk}F} \leq \mu(\ker_{\text{Coh}(S)}\varphi) \leq -y,$$

since $H \cdot c_1(\text{coker}_{\text{Coh}(S)}\varphi) \geq 0$, but $\text{rk}F < 0$. So, $-\mu(E) \geq y$, but this places $I(E)$ above (x, y) , so it must be above the parabola. $\square$

Thus, the heights of $I(F)$, $I(E)$ give lower and upper bounds respectively on the height for this wall, so the height ratio is bounded above by

$$\frac{-\mu(E)}{-\mu(F)} = \frac{r}{r-1},$$

which among integer values $r > 1$ is maximized at $r = 2$ to get a maximum height ratio of 2. Thus, to prevent high rank walls from destabilizing I_Z too early, we just need to require

$$\frac{r}{r-1} < \frac{C^2}{2d},$$

which is to require

$$C^2 > 4d.$$

2.3.2 Rank 1 Walls

Now, assume $\text{rk}F = r = 1$. First, we characterize such sheaves F .

Lemma 2.10. A rank 1 sheaf on a smooth projective surface is of the form

$$\mathcal{O}(-E) \otimes I_W.$$

Proof. To see this, notice that $F^{\vee\vee}$ is a line bundle by virtue of being reflexive rank 1 on a smooth variety, and the usual evaluation map $s \mapsto \text{eval}_s$ is injective in $\text{Coh}(S)$, giving us the short exact sequence

$$0 \rightarrow F \xrightarrow{e} F^{\vee\vee} \rightarrow \text{cokere} \rightarrow 0.$$

Twisting by the dual $F^\vee$, we get

$$0 \rightarrow F \otimes F^\vee \rightarrow \mathcal{O}_S \rightarrow \text{cokere} \otimes F^\vee \rightarrow 0,$$

which expresses F as a twist of an ideal sheaf (being a subsheaf of $\mathcal{O}_S$), and the Chern invariants will force it to be the ideal sheaf of a dimension 0 scheme. $\square$

Next, when analyze stability, we will be interested in morphisms $I_W \rightarrow I_Z$, so we prove this next lemma.

is this even helpful here?

Lemma 2.11. Let Z, W be closed subschemes of codimension at least 2 in a smooth projective variety X . Then, $\text{Hom}(I_W, I_Z)$ is $\mathbb{C}$ (spanned by inclusion) if $Z \subseteq W$ and 0 otherwise.

Proof. Let $\varphi : I_W \rightarrow I_Z$. The strategy is to analyze double duals. First, we argue that $I_W^{\vee\vee}, I_Z^{\vee\vee} \cong \mathcal{O}_X$ by applying $\mathcal{H}om_{\mathcal{O}_X}(-, \mathcal{O}_X)$ to, for example,

$$0 \rightarrow I_W \rightarrow \mathcal{O}_X \rightarrow \mathcal{O}_W \rightarrow 0$$

and trying to get vanishing

$$\mathcal{H}om_{\mathcal{O}_X}(\mathcal{O}_W, \mathcal{O}_X), \mathcal{E}xt^1(\mathcal{O}_W, \mathcal{O}_X) = 0.$$

First, $\mathcal{H}om$ vanishes because there is no nonzero assignment of any $1 \in \mathcal{O}_W(U)$ to a section of $\mathcal{O}_X$ due to torsion. Next, $\mathcal{E}xt^1_{\mathcal{O}_X}(\mathcal{O}_W, \mathcal{O}_X)$ can be computed by the spectral sequence (for the composition of functors $R\Gamma$ and $R\mathcal{H}om$)

$$H^p(X, \mathcal{E}xt^q_{\mathcal{O}_X}(\mathcal{O}_W, \mathcal{O}_X)) \Rightarrow \text{Ext}^{p+q}(\mathcal{O}_W, \mathcal{O}_X),$$

where how does this get vanishing? I might need something else.

Then, we form a commutative diagram

$$\begin{array}{ccc} I_W & \xrightarrow{\varphi} & I_Z \\ \downarrow & & \downarrow \\ I_W^{\vee\vee} & \xrightarrow{\varphi^{\vee\vee}} & I_Z^{\vee\vee} \end{array}$$

where the vertical maps send $s \mapsto \text{eval}_s$. In fancier language, $(-)^{\vee\vee}$ is an endofunctor, and $s \mapsto \text{eval}_s$ is a natural transformation from identity.

Now, the bottom morphism in our commutative diagram has to be multiplication by a constant $c \in \mathbb{C}$, and a nonzero c also forces the top morphism $I_W \rightarrow I_Z$ uniquely to be c multiplied by inclusion (in particular showing $Z \subseteq W$).

(from this math overflow post)

□

prove something like this?

Lemma 2.12. A morphism $\mathcal{O}_S(-E) \otimes I_W \rightarrow I_Z$ is the data of a section of $(I_Z : I_W)(E)$.

Proof. TODO/

□

Now, that we have a grip on the players, we try to handle stability. Our task is to rule out sheaves $\mathcal{O}_S(-E) \otimes I_W$ before the numerical wall through $I(I_Z)$ and $I(\mathcal{O}_S(-C)[1])$.

By lemma 2.12, if we have already ruled out destabilizers of lower length schemes Z' than Z , then we can assume $W = \emptyset$, since a morphism $\mathcal{O}_S(-E) \otimes I_W \rightarrow I_Z$ will correspond to a

$$\mathcal{O}_S(-E) \rightarrow (I_Z : I_W) = I_{Z'}$$

which is at least as “dangerous” as far as stability goes.

So, we are considering

References

- [1] Daniele Arcara and Aaron Bertram. *Reider’s Theorem and Thaddeus Pairs Revisited*. Apr. 22, 2009. DOI: 10.48550/arXiv.0904.3500. arXiv: 0904.3500[math]. URL: <http://arxiv.org/abs/0904.3500> (visited on 02/07/2026).
- [2] Aaron Bertram et al. *Two New Extensions of Reider’s Theorem on Algebraic Surfaces*. Apr. 23, 2026. DOI: 10.48550/arXiv.2604.21242. arXiv: 2604.21242[math]. URL: <http://arxiv.org/abs/2604.21242> (visited on 04/24/2026).